

Qixin (Lindsey) Deng

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EDUCATION

Northwestern University
Ph.D. in Computer Science

Evanston, IL
Sep 2026 – Jun 2030 (Expected)

Northwestern University
Master of Science in Electrical Engineering
GPA: 3.86/4.00

Evanston, IL
Sep 2024 – Jun 2026

Core Courses: Machine Learning, Deep Learning, Digital Signal Processing, Statistical Pattern Recognition

University of Rochester

Rochester, NY

Bachelor of Science in Audio and Music Engineering | Minor in Computer Science
GPA: 3.96/4.00 | Magna Cum Laude | Highest Distinction

Aug 2021 – May 2024

Core Courses: Audio Signal Processing, Audio Software Development, Computer Audition, Musical Acoustics

RESEARCH EXPERIENCE

Interactive Audio Lab, Northwestern University

Graduate Student Researcher

Sep 2024 – Present

- Advisor: Bryan Pardo and Thrasyvoulos N. Pappas

Multimodal Art Projection + Hong Kong University of Science and Technology

Research Collaborator

Sep 2023 – Jul 2025

- Advisor: Wei Xue

University of Rochester

Undergraduate Student Researcher

Jan 2024 – Apr 2024

- Advisor: Michael Heilemann

PUBLICATIONS

1. **Q. Deng**, B. Pardo, T. N. Pappas. Do Joint Language-Audio Embeddings Encode Perceptual Timbre Semantics?. In *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2026
2. **Q. Deng**, Q. Yang, R. Yuan, Y. Huang, Y. Wang, X. Liu, Z. Tian, J. Pan, G. Zhang, H. Lin, Y. Li, Y. Ma, J. Fu, C. Lin, E. Benetos, W. Wang, G. Xia, W. Xue, Y. Guo. ComposerX: Multi-Agent Symbolic Music Composition with LLMs. In *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR)*, 2024
3. F. Cwitkowitz, C. Benetatos, **Q. Deng**, H. Yu, N. Pruyne, P. O'Reilly, H. Flores Garcia, Z. Duan, B. Pardo. HARP 3.0: Generalizing I/O and API Support for Machine Learning in Digital Audio Workstations. In *NeurIPS 2025 Workshop on AI for Music*, 2025
4. C. Hao, R. Yuan, J. Yao, **Q. Deng**, X. Bai, Y. Wang, W. Xue, L. Xie. SongFormer: Scaling Music Structure Analysis with Heterogeneous Supervision. In *Proceedings of the 2026 IEEE International Conference on Multimedia and Expo (ICME)*, 2026

TEACHING EXPERIENCE

ECE 113: Circuits & Signals, University of Rochester

Workshop Leader

Spring 2023

AWARDS & HONORS

Cognitive Science Fellowship

Northwestern University

2026

NeurIPS AI for Music Workshop Student Grant*NeurIPS AI4Music*

2025

Phi Beta Kappa Academic Honor Society*ECE Department, University of Rochester*

2024

Tau Beta Pi Engineering Honor Society*ECE Department, University of Rochester*

2023

Whipple Science and Research Scholarship (\$12,000 per year)*University of Rochester*

2021 – 2024

SERVICE

Reviewer

2026

International Society for Music Information Retrieval (ISMIR)

SKILLS

Programming Languages – Python, C/C++, MATLAB**Audio Programming Languages** – Faust, Max/MSP**Tools** – NumPy, PyTorch, LaTeX, Git, JUCE**Acoustical Measurement** – CLIO, Room EQ Wizard**Hardware Design** – LTspice, KiCad**Audio Engineering** – Studio recording, mixing, mastering in Logic Pro and Pro Tools**Musical Instruments** – Piano, guitar